

Exercices on session *Getting started with R* (2/2)

Matrices

We are going to use a matrix called *occ* to get data on occurrence of species in sites (1 if the species is present, and 0 if it is absent).

- Create the following matrix and check that we have the structure we want.

```
occ <- matrix(c(1, 1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 0, 0, 1, 1, 0, 1),
              4,
              5)
colnames(occ) <- c("Sp1", "Sp2", "Sp3", "Sp4", "Sp5")
rownames(occ) <- c("site1", "site2", "site3", "site4")
occ
```

	Sp1	Sp2	Sp3	Sp4	Sp5
site1	1	0	1	0	1
site2	1	1	1	1	1
site3	1	0	0	0	0
site4	1	0	1	0	1

```
str(occ)
```

- Calculate the number of species in each site.

```
rowSums(occ)
```

- Calculate the number of sites in which each species occurs.

```
colSums(occ)
```

- Transpose the matrix to get the following:

```
t(occ)
```

	site1	site2	site3	site4
Sp1	1	1	1	1
Sp2	0	1	0	0
Sp3	1	1	0	1
Sp4	0	1	0	0
Sp5	1	1	0	1

Lists

- Create the following list:

```
v <- c("site1" = "DF", "site2" = "GO", "site3" = "GO", "site4" = "MT")
dt <- list(occurrence = occ,
           site_state = v)
dt
```

```
$occurrence
      Sp1 Sp2 Sp3 Sp4 Sp5
site1   1  0  1  0  1
site2   1  1  1  1  1
site3   1  0  0  0  0
site4   1  0  1  0  1
```

```
$site_state
site1 site2 site3 site4
"DF"  "GO"  "GO"  "MT"
```

Operators and Functions

- Calculate the square root of 25

```
sqrt(25)
```

- Check that the exponential of the logarithm of 1 is equal to 1. *NB: If you try to do it for another number, you will get an unexpected result due to rounding.*

```
exp(log(1)) == 1  
  
# for other values  
round(exp(log(3))) == 3
```

- Check that 2 is between 0 and 10 (there are different ways to do it)

```
2 %in% 0:10  
# or  
2 <= 10 & 2 >= 0
```

- Get the absolute value of each element of the following vector *vec*:

```
vec <- c(-4, 5, 8, -9, -3)  
vec
```

```
[1] -4  5  8 -9 -3
```

```
abs(vec)
```

- Get the range of values in *vec*

```
range(vec)
```

- Calculate the basal area of a tree of diameter 25 cm (the basal area is the area of a circle that has the diameter of the consider tree), in m²

```
pi * (25/200)^2
```